FINANCIAL ENGINEERING

KBP FFÚ & KSTP FIS

Case Studies in Capital Markets and Banking

Course introduction

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EVROPSKÁ UNIE
Evropské strukturální a investiční fondy
Operační program Výzkum, vývoj a vzdělávání

MINISTERSTVO ŠKOLSTVÍ, MLÁDEŽE A TĚLOVÝCHOVY

Course organization

- **Requirements:**
  - Course projects (50%)
  - Final test (50%)
- **Grading:**
  - Excellent (1) (90-100%)
  - Very good (2) (75-89%)
  - Good (3) (60-74%)
  - Insufficient (possible repair) (4+) (50-59%)
  - Insufficient (no repair) (4) (0-49%)

Literature

Type	Author	Title	Year
RQ	WITZANY, J.	Financial Derivatives. Valuation, Hedging and Risk Management	2013
RE	WITZANY, J.	International Financial Markets	2012
RE	HULL, J.	Options, futures and other derivatives	2015
RE	HULL, J.	Risk management and financial institutions	2012
RE	BODIE, Z. -- KANE, A. -- MARCUS, A J.	Investments	2009
RE	EAKINS, S G. -- MISHKIN, F S.	Financial markets and institutions	2018
RE	WILMOTT, P.	Paul Wilmott on quantitative finance. Volume two	2006
RE	RAČEV, S.	Financial econometrics : from basics to advanced modeling techniques	2007

Course topics

1. **Introduction to R** (2 lectures)
   - R data types and operations, vectors and sequences, conditions and cycles, user build functions, matrices and matrix operations, lists and data arrays, charts and figures, R packages
2. **Stock markets** (4 lectures)
   - Modern Portfolio Theory, CAPM model, factor models, market risk premium, equity premium puzzle, Active vs. passive investing, Efficient Market Hypothesis, fundamental and technical analysis, tests of market efficiency, market anomalies, statistical arbitrage, global macro strategies
3. **Bond markets** (1 lecture)
   - Risks influencing a bond investments, duration and convexity, interest rate term structure, models of the yield curve, liquidity risk premium, forward interest rates, riding of the yield curve
4. **Foreign exchange markets** (1 lecture)
   - Currency risk, absolute and relative purchasing power parity, real exchange rate, covered and uncovered interest rate parity, exchange rate factor models
5. **Futures markets** (1 lecture)
   - Investing with futures, differences between forwards and futures, hedging a portfolio of stocks and bonds against market risk, speculators and arbitrageurs equilibrium, equilibrium futures price, normal vs. inverted futures term structure, contango vs. normal backwardation
6. **Option markets** (4 lectures)
   - Put-call parity, Black-Scholes model, binomial trees, Monte-Carlo method, Finite difference method, option Greeks, hedging of an option portfolio, option investing strategies, implied volatility, volatility surface, Model-Free volatility, volatility risk premium, alternative stochastic models, CEV model, IVF model, Heston model, SABR model, pricing of exotic options